

Compute Express Link Specification

Revision 3.2, Version 1.0

第 1 章 导论 / Chapter 1.0 Introduction

中英对照译本（非官方）

Chinese-English Bilingual Edition (Unofficial)

源文档：CXL Specification Rev 3.2, Version 1.0 (2024-10-02) Evaluation Copy — Chapter 1
本译本仅供学习参考，术语以原文为准。插图保留英文原图。

1.0 Introduction / 导论

1.1 Audience / 读者对象

EN

The information in this document is intended for anyone designing or architecting any hardware or software associated with Compute Express Link (CXL) or Flex Bus.

中文

本文档中的信息面向任何设计与架构与 Compute Express Link (CXL) 或 Flex Bus 相关的硬件或软件的人员。

1.2 Terminology/Acronyms / 术语与缩略语

EN

Refer to PCIe Base Specification for additional terminology and acronym definitions beyond those listed in Table 1-1.

中文

表 1-1 未列出的其他术语与缩略语定义，请参阅 PCIe 基础规范。

Table 1-1. Terminology/Acronyms / 表 1-1 术语与缩略语

下表按英文字母序给出原文定义与中文译文。专有缩写译文中保留英文形式。

AAD	Additional Authentication Data - data that is integrity protected but not encrypted	附加认证数据 (Additional Authentication Data) ——完整性受保护但不加密的数据
Accelerator / Acc	Devices that may be used by software running on Host processors to offload or perform any type of compute or I/O task. Examples of accelerators include programmable agents (e.g., GPU/GPGPU), fixed- function agents, or reconfigurable agents such as FPGAs.	加速器：可由主机处理器上运行的软件用于卸载或执行任意类型计算或 I/O 任务的设备。加速器示例包括可编程代理（如 GPU/GPGPU）、固定功能代理，或可重构代理（如 FPGA）。
ACL	Access Control List	访问控制列表 (Access Control List)
ACPI	Advanced Configuration and Power Interface	高级配置与电源接口 (Advanced Configuration and Power Interface)
ACS	Access Control Services as defined in PCIe Base Specification	访问控制服务 (Access Control Services)，如 PCIe 基础规范所定义
ADF	All-Data Flit	全数据 Flit (All-Data Flit)
AER	Advanced Error Reporting as defined in PCIe Base Specification	高级错误报告 (Advanced Error Reporting)，如 PCIe 基础规范所定义
AES-GCM	Advanced Encryption Standard-Galois Counter Mode as defined in NIST* publication [AES-GCM]	高级加密标准-伽罗瓦计数器模式 (AES-GCM)，如 NIST* 出版物 [AES-GCM] 所定义
AIC	Add In Card	附加卡 (Add In Card)
Ak	Acknowledgment (bit/field name)	确认 (Acknowledgment，位/字段名)
ALMP	ARB/MUX Link Management Packet	ARB/MUX 链路管理包 (ARB/MUX Link Management Packet)
AP	Auto Precharge	自动预充电 (Auto Precharge)
ARB/MUX	Arbiter/Multiplexer	仲裁器/复用器 (Arbiter/Multiplexer)
ARI	Alternate Routing ID as defined in PCIe Base Specification.	备用路由 ID (Alternate Routing ID)，如 PCIe 基础规范所定义。
ASI	Advanced Switching Interconnect	高级交换互连 (Advanced Switching Interconnect)
ASL	ACPI Source Language as defined in ACPI Specification	ACPI 源语言 (ACPI Source Language)，如 ACPI 规范所定义
ASPM	Active State Power Management	主动状态电源管理 (Active State Power Management)

ATS	Address Translation Services as defined in PCIe Base Specification	地址转换服务（Address Translation Services），如 PCIe 基础规范所定义
Attestation	Process of providing a digital signature for a set of measurements from a device and verifying those signatures and measurements on a host.	证明：为设备的一组度量提供数字签名，并在主机上验证这些签名与度量的过程。
Authentication	Process of determining whether an entity is who or what it claims to be.	认证：确定某个实体是否为其声称的身份或对象的过程。
BAR	Base Address Register as defined in PCIe Base Specification	基址寄存器（Base Address Register），如 PCIe 基础规范所定义
_BBN	Base Bus Number as defined in ACPI Specification	基总线号（Base Bus Number），如 ACPI 规范所定义
BCC	Base Class Code as defined in PCIe Base Specification	基类代码（Base Class Code），如 PCIe 基础规范所定义
BDF	Bus Device Function	总线/设备/功能（Bus Device Function）
BE	Byte Enable as defined in PCIe Base Specification	字节使能（Byte Enable），如 PCIe 基础规范所定义
BEI	BAR Equivalent Indicator	BAR 等价指示符（BAR Equivalent Indicator）
BEP	Byte-Enables Present	存在字节使能（Byte-Enables Present）
BI	Back-Invalidate	反向使无效（Back-Invalidate）
Bias Flip	Bias refers to coherence tracking of HDM-D* memory regions by the owning device to indicate that the host may have a cache copy. The Bias Flip is a process used to change the bias state that indicates the host has a cached copy of the line (Bias=Host) by invalidating the cache state for the corresponding address(es) in the host such that the device has exclusive access (Bias=Device).	Bias（偏置）指拥有设备对 HDM-D* 内存区域进行一致性跟踪，以指示主机可能持有缓存副本。Bias Flip（偏置翻转）是一种用于改变偏置状态的过程：当状态指示主机持有该行的缓存副本（Bias=Host）时，通过使主机中对应地址的缓存状态无效，使设备获得独占访问（Bias=Device）。
BIR	BAR Indicator Register as defined in PCIe Base Specification	BAR 指示寄存器（BAR Indicator Register），如 PCIe 基础规范所定义
BIRsp	Back-Invalidate Response	反向使无效响应（Back-Invalidate Response）
BISnp	Back-Invalidate Snoop	反向使无效监听（Back-Invalidate Snoop）
BMC	Baseboard Management Controller	基板管理控制器（Baseboard Management Controller）
BME	Bus Master Enable	总线主控使能（Bus Master Enable）
BW	Bandwidth	带宽（Bandwidth）
CA	Certificate Authority; Congestion Avoidance	证书颁发机构（Certificate Authority）；拥塞避免（Congestion Avoidance）
CAM	Content Addressable Memory	内容可寻址存储器（Content Addressable Memory）
CAS	Column Address Strobe	列地址选通（Column Address Strobe）
_CBR	CXL Host Bridge Register Info as defined in ACPI Specification	CXL 主机桥寄存器信息，如 ACPI 规范所定义
CCI	Component Command Interface	组件命令接口（Component Command Interface）

CCID	CXL Cache ID	CXL Cache ID
CDAT	Coherent Device Attribute Table, a table that describes performance characteristics of a CXL device or a CXL switch.	一致性设备属性表 (Coherent Device Attribute Table)，用于描述 CXL 设备或 CXL 交换机性能特征的表。
CDL	CXL DevLoad	CXL DevLoad
CEDT	CXL Early Discovery Table	CXL 早期发现表 (CXL Early Discovery Table)
CEL	Command Effects Log	命令效应日志 (Command Effects Log)
CFMWS	CXL Fixed Memory Window Structure	CXL 固定内存窗口结构 (CXL Fixed Memory Window Structure)
CHBCR	CXL Host Bridge Component Registers	CXL 主机桥组件寄存器 (CXL Host Bridge Component Registers)
CHBS	CXL Host Bridge Structure	CXL 主机桥结构 (CXL Host Bridge Structure)
_CID	Compatible ID as defined in ACPI Specification	兼容 ID (Compatible ID)，如 ACPI 规范所定义
CIE	Correctable Internal Error	可纠正内部错误 (Correctable Internal Error)
CIKMA	CXL.cachemem IDE Key Management Agent	CXL.cachemem IDE 密钥管理代理 (CXL.cachemem IDE Key Management Agent)
CKID	Context Key IDentifier passed in the protocol flit for identifying security keys utilized for memory encryption using TSP.	上下文密钥标识符 (Context Key IDentifier)，在协议 flit 中传递，用于标识利用 TSP 进行内存加密所使用的安全密钥。
CMA	Component Measurement and Authentication as defined in PCIe Base Specification	组件度量与认证 (Component Measurement and Authentication)，如 PCIe 基础规范所定义
Coh	Coherency	一致性 (Coherency)
Cold reset	As defined in PCIe Base Specification	冷复位，如 PCIe 基础规范所定义
Comprehensive Trust	Security model in which every device available to the TEE is presumed to be trusted by all TEEs in the system.	全面信任：一种安全模型，其中 TEE 可用的每个设备都被假定为受系统中所有 TEE 信任。
CPMU	CXL Performance Monitoring Unit	CXL 性能监控单元 (CXL Performance Monitoring Unit)
CRD	Credit Return(ed)	信用返回 (Credit Return(ed))
CQID	Command Queue ID	命令队列 ID (Command Queue ID)
CSDS	CXL System Description Structure	CXL 系统描述结构 (CXL System Description Structure)
CSR	Configuration Space register	配置空间寄存器 (Configuration Space register)
CT	Crypto Timeout	加密超时 (Crypto Timeout)
CVME	Corrected Volatile Memory Error, a corrected error associated with volatile memory.	已纠正易失性内存错误 (Corrected Volatile Memory Error)，与易失性内存相关的已纠正错误。
CXIMS	CXL XOR Interleave Math Structure	CXL XOR 交织数学结构 (CXL XOR Interleave Math Structure)
CXL	Compute Express Link, a low-latency, high-bandwidth link that supports dynamic protocol muxing of coherency, memory access, and I/O protocols, thus enabling attachment of coherent accelerators or memory devices.	Compute Express Link, 一种低延迟、高带宽链路，支持一致性、内存访问与 I/O 协议的动态协议复用，从而支持连接一致性加速器或内存设备。
CXL.cache	Agent coherency protocol that supports device caching of Host memory.	代理一致性协议，支持设备缓存主机内存。
CXL.cachemem	CXL.cache/CXL.mem	CXL.cache/CXL.mem
CXL.io	PCIe-based non-coherent I/O protocol with enhancements for accelerator support.	基于 PCIe 的非一致性 I/O 协议，并针对加速器支持进行了增强。
CXL.mem	Memory access protocol that supports device-attached memory.	内存访问协议，支持设备附属内存。

CXL Memory Device	CXL device with a specific Class Code as defined in Section 8.1.12.1.	具有特定类代码的 CXL 设备，如第 8.1.12.1 节所定义。
D2H	Device to Host	设备到主机（Device to Host）
DAPM	Deepest Allowable Power Management state	允许的最深电源管理状态（Deepest Allowable Power Management state）
DC	Dynamic Capacity	动态容量（Dynamic Capacity）
DCD	Dynamic Capacity Device	动态容量设备（Dynamic Capacity Device）
DCOH	Device Coherency agent on the device that is responsible for resolving coherency with respect to device caches and managing Bias states.	设备一致性代理（Device Coherency agent）：设备上负责相对设备缓存解决一致性并管理 Bias 状态的代理。
DDR	Double Data Rate	双倍数据速率（Double Data Rate）
DH	Data Header	数据头（Data Header）
DHSW-FM	Dual Host, Fabric Managed, Switch Attached SLD EP	双主机、结构管理、交换机连接的 SLD 端点
DHSW-FM-MLD	Dual Host, Fabric Managed, Switch Attached MLD EP	双主机、结构管理、交换机连接的 MLD 端点
DLLP	Data Link Layer Packet as defined in PCIe Base Specification	数据链路层包（Data Link Layer Packet），如 PCIe 基础规范所定义

DM	Data Mask	数据掩码 (Data Mask)
DMP	Device Media Partition	设备介质分区 (Device Media Partition)
DMTF	Distributed Management Task Force	分布式管理任务组 (Distributed Management Task Force)
DOE	Data Object Exchange as defined in PCIe Base Specification	数据对象交换 (Data Object Exchange), 如 PCIe 基础规范所定义
Domain	Set of host ports and devices within a single coherent HPA space	单一一致性 HPA 空间内的一组主机端口与设备
Downstream ES	Within the context of a single VH, an Edge Switch other than the Host ES.	在单个 VH 的上下文中, 除 Host ES 以外的边缘交换机。
Downstream Port	Physical port that can be a root port or a downstream switch port or an RCH Downstream Port	物理端口, 可以是根端口、下游交换机端口或 RCH 下游端口
DPA	Device Physical Address. DPA forms a device-scoped flat address space. An LD-FAM device presents a distinct DPA space per LD. A G-FAM device presents the same DPA space to all hosts. The CXL HDM decoders or GFD decoders map HPA into DPA space.	设备物理地址 (Device Physical Address)。DPA 构成设备范围内的平坦地址空间。LD-FAM 设备为每个 LD 呈现不同的 DPA 空间。G-FAM 设备向所有主机呈现相同的 DPA 空间。CXL HDM 解码器或 GFD 解码器将 HPA 映射到 DPA 空间。
DPC	Downstream Port Containment as defined in PCIe Base Specification	下游端口遏制 (Downstream Port Containment), 如 PCIe 基础规范所定义
DPID	Destination PID	目的 PID (Destination PID)
DRS	Data Response	数据响应 (Data Response)
DRT	DPID Routing Table	DPID 路由表 (DPID Routing Table)
DSAR	Downstream Acceptance Rules	下游接受规则 (Downstream Acceptance Rules)
_DSM	Device Specific Method as defined in ACPI Specification	设备特定方法 (Device Specific Method), 如 ACPI 规范所定义
DSM	Device Security Manager. A logical entity in a device that can be admitted into the TCB for a TVM and enforces security policies on the device.	设备安全管理器 (Device Security Manager)。设备中的逻辑实体, 可被纳入 TVM 的 TCB, 并在设备上强制执行安全策略。
DSMAD	Device Scoped Memory Affinity Domain as defined in Coherent Device Attribute Table (CDAT) Specification	设备范围内内存亲和域, 如一致性设备属性表 (CDAT) 规范所定义
DSMAS	Device Scoped Memory Affinity Structure as defined in Coherent Device Attribute Table (CDAT) Specification	设备范围内内存亲和结构, 如一致性设备属性表 (CDAT) 规范所定义
DSP	Downstream Switch Port	下游交换机端口 (Downstream Switch Port)
DTLB	Data Translation Lookaside Buffer	数据转换旁路缓冲 (Data Translation Lookaside Buffer)
DTRCS	Device Tracked Requestor Cache State – The requestor cache coherency state tracked by the device.	设备跟踪的请求方缓存状态 (Device Tracked Requestor Cache State) —— 由设备跟踪的请求方缓存一致性状态。
DUT	Device Under Test	被测设备 (Device Under Test)
DVSEC	Designated Vendor-Specific Extended Capability as defined in PCIe Base Specification	指定厂商特定扩展能力 (Designated Vendor-Specific Extended Capability), 如 PCIe 基础规范所定义
ECC	Error-correcting Code	纠错码 (Error-correcting Code)
ECN	Engineering Change Notice	工程变更通知 (Engineering Change Notice)
ECS	Error Check Scrub	错误检查清洗 (Error Check Scrub)
ECRC	End-to-End CRC	端到端 CRC (End-to-End CRC)
EDB	End Bad as defined in PCIe Base Specification	坏结束 (End Bad), 如 PCIe 基础规范所定义
Edge DSP	PBR downstream switch port that connects to a device (including a GFD) or an HBR USP	连接到设备 (含 GFD) 或 HBR USP 的 PBR 下游交换机端口

Edge Port	PBR switch port suitable for connecting to an RP, device, or HBR USP.	适于连接 RP、设备或 HBR USP 的 PBR 交换机端口。
Edge Switch (ES)	Within the context of a single VH, a PBR switch that contains one or more Edge Ports.	在单个 VH 的上下文中，包含一个或多个边缘端口的 PBR 交换机。
Edge USP	PBR upstream switch port that connects to a root port	连接到根端口的 PBR 上游交换机端口
eDPC	Enhanced Downstream Port Control as defined in PCIe Base Specification	增强下游端口控制（Enhanced Downstream Port Control），如 PCIe 基础规范所定义
EDS	End of Data Stream	数据流结束（End of Data Stream）
EFN	Event Firmware Notification	事件固件通知（Event Firmware Notification）
EIEOS	Electrical Idle Exit Ordered Set as defined in PCIe Base Specification	电气空闲退出有序集，如 PCIe 基础规范所定义
EIOS	Electrical Idle Ordered Set as defined in PCIe Base Specification	电气空闲有序集，如 PCIe 基础规范所定义
EIOSQ	EIOS Sequence as defined in PCIe Base Specification	EIOS 序列，如 PCIe 基础规范所定义
EMD	Extended Metadata	扩展元数据（Extended Metadata）
EMV	Extended MetaValue	扩展元值（Extended MetaValue）
ENIW	Encoded Number of Interleave Ways	编码交织路数（Encoded Number of Interleave Ways）

EP	Endpoint	端点 (Endpoint)
eRCD	Exclusive Restricted CXL Device (formerly known as CXL 1.1 only Device). eRCD is a CXL device component that can operate only in RCD mode.	独占受限 CXL 设备 (Exclusive Restricted CXL Device, 原称仅 CXL 1.1 设备)。eRCD 是只能在 RCD 模式下运行的 CXL 设备组件。
eRCH	Exclusive Restricted CXL Host (formerly known as CXL 1.1 only host). eRCH is a CXL host component that can operate only in RCD mode.	独占受限 CXL 主机 (Exclusive Restricted CXL Host, 原称仅 CXL 1.1 主机)。eRCH 是只能在 RCD 模式下运行的 CXL 主机组件。
evPPB	Egress vPPB	出口 vPPB (Egress vPPB)
Explicit Access Control	TE State ownership tracking where the state change occurs explicitly utilizing a TE State change message that is independent of each memory transaction.	显式访问控制: TE 状态所有权跟踪方式, 状态变更通过独立于每次内存事务的 TE 状态变更消息显式发生。
Fabric Port (FPort)	PBR switch port that connects to another PBR switch port.	连接到另一个 PBR 交换机端口的 PBR 交换机端口。
FAM	Fabric-Attached Memory. HDM within a Type 2 or Type 3 device that can be made accessible to multiple hosts concurrently. Each HDM region can either be pooled (dedicated to a single host) or shared (accessible concurrently by multiple hosts).	结构附属内存 (Fabric-Attached Memory)。Type 2 或 Type 3 设备中的 HDM, 可同时被多个主机访问。每个 HDM 区域既可池化 (专用于单个主机), 也可共享 (由多个主机并发访问)。
FAST	Fabric Address Segment Table	结构地址段表 (Fabric Address Segment Table)
FC	Flow Control	流控 (Flow Control)
FCBP	Flow Control Backpressured	流控背压 (Flow Control Backpressured)
FEC	Forwarded Error Correction	前向纠错 (Forwarded Error Correction)
Flex Bus	A flexible high-speed port that is configured to support either PCIe or CXL.	可配置为支持 PCIe 或 CXL 的灵活高速端口。
Flex Bus.CXL	CXL protocol over a Flex Bus interconnect.	在 Flex Bus 互连上运行的 CXL 协议。
Flit	Link Layer Unit of Transfer	链路层传输单元 (Link Layer Unit of Transfer)
FLR	Function Level Reset	功能级复位 (Function Level Reset)
FM	The Fabric Manager is an entity separate from the switch or host firmware that controls aspects of the system related to binding and management of pooled ports and devices.	结构管理器 (Fabric Manager) 是独立于交换机或主机固件的实体, 控制与池化端口和设备的绑定及管理相关的系统方面。
FMLD	Fabric Manager-owned Logical Device	结构管理器拥有的逻辑设备 (Fabric Manager-owned Logical Device)
FM-owned PPB	Link that contains traffic from multiple VCSs or an unbound physical port.	包含来自多个 VCS 的流量或未绑定物理端口的链路。
Fundamental Reset	As defined in PCIe Base Specification	基本复位, 如 PCIe 基础规范所定义
FW	Firmware	固件 (Firmware)
GAM	GFD Async Message	GFD 异步消息 (GFD Async Message)
GAE	Global Memory Access Endpoint	全局内存访问端点 (Global Memory Access Endpoint)
GDT	GFD Decoder Table	GFD 解码器表 (GFD Decoder Table)
G-FAM	Global FAM. Highly scalable form of FAM that is presented to hosts without using Logical Devices (LDs). Like LD-FAM, G-FAM presents HDM that can be pooled or shared.	全局 FAM。不使用逻辑设备 (LD) 即可向主机呈现的高度可扩展 FAM 形式。与 LD-FAM 类似, G-FAM 呈现可池化或可共享的 HDM。
GFD	G-FAM device	G-FAM 设备
GFD decoder	HPA-to-DPA translation mechanism inside a GFD	GFD 内部的 HPA 到 DPA 转换机制
GI	Generic Affinity	通用亲和 (Generic Affinity)

GIM	Global Integrated Memory	全局集成内存 (Global Integrated Memory)
GMV	Global Memory Mapping Vector	全局内存映射向量 (Global Memory Mapping Vector)
GO	Global Observation. This is used in the context of coherence protocol as a message to know when data is guaranteed to be observed by all agents in the coherence domain for either a read or a write.	全局可见 (Global Observation)。在一致性协议上下文中用作消息，用于获知数据在读或写时保证可被一致性域中所有代理观察到的时机。
GPF	Global Persistent Flush	全局持久刷新 (Global Persistent Flush)
H2D	Host to Device	主机到设备 (Host to Device)
HA	Home Agent. This is the agent on the host that is responsible for resolving system wide coherency for a given address.	Home Agent。主机上负责针对给定地址解决系统范围一致性的代理。
HBIG	Host Bridge Interleave Granularity	主机桥交织粒度 (Host Bridge Interleave Granularity)
HBM	High-Bandwidth Memory	高带宽内存 (High-Bandwidth Memory)
HBR	Hierarchy Based Routing	基于层级的路由 (Hierarchy Based Routing)
HBR link	Link operating in a mode, where all flits are in HBR format	以所有 flit 均为 HBR 格式的模式运行的链路
HBR switch	Switch that supports only HBR	仅支持 HBR 的交换机
HDM	Host-managed Device Memory. Device-attached memory that is mapped to system coherent address space and accessible to the Host using standard write-back semantics. Memory located on a CXL device can be mapped as either HDM or PDM.	主机管理的设备内存 (Host-managed Device Memory)。映射到系统一致性地址空间、主机可使用标准写回语义访问的设备附属内存。位于 CXL 设备上的内存可映射为 HDM 或 PDM。
HDM-H	Host-only Coherent HDM region type. Used only for Type 3 Devices.	仅主机一致性的 HDM 区域类型。仅用于 Type 3 设备。

HDM-D	Device Coherent HDM region type. Used only for Type 2 devices that rely on CXL.cache to manage coherence with the host.	设备一致性 HDM 区域类型。仅用于依赖 CXL.cache 与主机管理一致性的 Type 2 设备。
HDM-DB	Device Coherent using Back-Invalidate HDM region type. Can be used by Type 2 devices or Type 3 devices.	使用反向使无效的设备一致性 HDM 区域类型。可由 Type 2 或 Type 3 设备使用。
_HID	Hardware ID as defined in ACPI Specification	硬件 ID (Hardware ID) , 如 ACPI 规范所定义
HMAT	Heterogeneous Memory Attribute Table as defined in ACPI Specification	异构内存属性表 (Heterogeneous Memory Attribute Table) , 如 ACPI 规范所定义
Host ES	Within the context of a single VH, the single Edge Switch connected to the RP.	在单个 VH 的上下文中, 连接到 RP 的唯一边缘交换机。
Hot Reset	As defined in PCIe Base Specification	热复位, 如 PCIe 基础规范所定义
HPA	Host Physical Address	主机物理地址 (Host Physical Address)
HPC	High-Performance Computing	高性能计算 (High-Performance Computing)
hPPR	Hard Post Package Repair	硬封装后修复 (Hard Post Package Repair)
HW	Hardware	硬件 (Hardware)
IDE	Integrity and Data Encryption	完整性与数据加密 (Integrity and Data Encryption)
IDT	Interleave DPID Table	交织 DPID 表 (Interleave DPID Table)
IG	Interleave Granularity	交织粒度 (Interleave Granularity)
IGB	Interleave Granularity in number of bytes	以字节数表示的交织粒度
Intermediate Fabric Switch	Within the context of a single VH, a PBR switch that forwards VH traffic but is not a Host ES or downstream ES.	在单个 VH 的上下文中, 转发 VH 流量但不是 Host ES 或下游 ES 的 PBR 交换机。
IOMMU	I/O Memory Management Unit as defined in PCIe Base Specification	I/O 内存管理单元, 如 PCIe 基础规范所定义
Implicit Access Control	TE State ownership tracking where the state change occurs implicitly as part of executing each memory transaction.	隐式访问控制: TE 状态所有权跟踪方式, 状态变更作为执行每次内存事务的一部分隐式发生。
Initiator	Defined by TSP as a host or accelerator device that issues TSP requests.	由 TSP 定义为发出 TSP 请求的主机或加速器设备。
IP2PM	Independent Power Manager to (Master) Power Manager, PM messages from the device to the host.	独立电源管理器到 (主) 电源管理器, 从设备到主机的 PM 消息。
ISL	Inter-Switch Link. PBR link that connects Fabric Ports.	交换机间链路 (Inter-Switch Link) 。连接结构端口的 PBR 链路。
ISP	Interleave Set Position	交织集位置 (Interleave Set Position)
IV	Initialization Vector	初始化向量 (Initialization Vector)
IW	Interleave Ways	交织路数 (Interleave Ways)
IWB	Implicit Writeback	隐式写回 (Implicit Writeback)
LAV	LDST Access Vector	LDST 访问向量 (LDST Access Vector)
LD	Logical Device. Entity that represents a CXL Endpoint that is bound to a VCS. An SLD contains one LD. An MLD contains multiple LDs.	逻辑设备 (Logical Device) 。表示绑定到 VCS 的 CXL 端点的实体。SLD 包含一个 LD。MLD 包含多个 LD。
LDST	LD-FAM Segment Table	LD-FAM 段表 (LD-FAM Segment Table)
LD-FAM	FAM that is presented to hosts via Logical Devices (LDs).	通过逻辑设备 (LD) 向主机呈现的 FAM。
Link Layer Clock	CXL.cachemem link layer clock rate usually defined by the flit rate during normal operation.	CXL.cachemem 链路层时钟速率, 通常由正常运行期间的 flit 速率定义。
LLC	Last Level Cache	末级缓存 (Last Level Cache)

LLCRD	Link Layer Credit	链路层信用 (Link Layer Credit)
LLCTRL	Link Layer Control	链路层控制 (Link Layer Control)
LLR	Link Layer Retry	链路层重试 (Link Layer Retry)
LLRB	Link Layer Retry Buffer	链路层重试缓冲 (Link Layer Retry Buffer)
LOpt	Latency-Optimized	延迟优化 (Latency-Optimized)
LRSM	Local Retry State Machine	本地重试状态机 (Local Retry State Machine)
LRU	Least recently used	最近最少使用 (Least recently used)
LSA	Label Storage Area	标签存储区 (Label Storage Area)
LSM	Link State Machine	链路状态机 (Link State Machine)
LTR	Latency Tolerance Reporting	延迟容限报告 (Latency Tolerance Reporting)

LTSSM	Link Training and Status State Machine as defined in PCIe Base Specification	链路训练与状态状态机，如 PCIe 基础规范所定义
LUN	Logical Unit Number	逻辑单元号 (Logical Unit Number)
M2S	Master to Subordinate	主到从 (Master to Subordinate)
MAC	Message Authentication Code; also referred to as Authentication Tag or Integrity value.	消息认证码 (Message Authentication Code) ；也称为认证标签或完整性值。
MAC epoch	Set of flits which are aggregated together for MAC computation.	为进行 MAC 计算而聚合在一起的一组 flit。
MB	Megabyte - 220 bytes (1,048,576 bytes)	兆字节 (Megabyte) ——2 ²⁰ 字节 (1,048,576 字节)
Mailbox	(No definition provided in source table.)	(源表未提供独立释义)
MC	Memory Controller	内存控制器 (Memory Controller)
MCA	Machine Check Architecture	机器检查架构 (Machine Check Architecture)
MCTP	Management Component Transport Protocol	管理组件传输协议 (Management Component Transport Protocol)
MDH	Multi-Data Header	多数据头 (Multi-Data Header)
Measurement	Representation of firmware/software or configuration data on a device.	设备上固件/软件或配置数据的表示。
MEC	Multiple Event Counting	多事件计数 (Multiple Event Counting)
MEFN	Memory Error Firmware Notification. An EFN that is used to report memory errors.	内存错误固件通知 (Memory Error Firmware Notification) 。用于报告内存错误的 EFN。
Memory Group	Set of DC blocks that can be accessed by the same set of requesters.	可由同一组请求者访问的一组 DC 块。
Memory Sparing	Repair function that replaces a portion of memory (the “spared memory”) with a portion of functional memory at that same DPA.	用同一 DPA 处的一部分功能内存替换一部分内存 (“备用内存”) 的修复功能。
MESI	Modified, Exclusive, Shared, and Invalid cache coherence protocol	Modified、Exclusive、Shared 与 Invalid 缓存一致性协议
MGT	Memory Group Table	内存组表 (Memory Group Table)
MH-MLD	Multi-headed MLD. CXL component that contains multiple CXL ports, each presenting an MLD or SLD. The ports must correctly operate when connected to any combination of common or different hosts. The FM API is used to configure each LD as well as the overall MH-MLD. Currently, MH-MLDs are architected only for Type 3 LDs. MH-MLDs are considered a specialized type of MLD and, as such, are subject to all functional and behavioral requirements of MLDs.	多头 MLD。包含多个 CXL 端口的 CXL 组件，每个端口呈现 MLD 或 SLD。这些端口在连接到任意组合的相同或不同主机时必须正确工作。使用 FM API 配置每个 LD 以及整体 MH-MLD。当前 MH-MLD 仅针对 Type 3 LD 进行架构定义。MH-MLD 被视为一种特殊类型的 MLD，因此须满足 MLD 的全部功能与行为要求。
MH-SLD	Multi-headed SLD. CXL component that contains multiple CXL ports, each presenting an SLD. The ports must correctly operate when connected to any combination of common or different hosts. The FM API is used to configure each SLD as well as the overall MH-SLD. Currently, MH-SLDs are architected only for Type 3 LDs.	多头 SLD。包含多个 CXL 端口的 CXL 组件，每个端口呈现 S LD。这些端口在连接到任意组合的相同或不同主机时必须正确工作。使用 FM API 配置每个 SLD 以及整体 MH-SLD。当前 MH-SLD 仅针对 Type 3 LD 进行架构定义。
ML	Machine Learning	机器学习 (Machine Learning)
MLD	Multi-Logical Device. CXL component that contains multiple LDs, out of which one LD is reserved for configuration via the FM API, and each remaining LD is suitable for assignment to a different host. Currently MLDs are architected only for Type 3 LDs.	多逻辑设备 (Multi-Logical Device) 。包含多个 LD 的 CXL 组件，其中一个 LD 保留用于通过 FM API 配置，其余每个 LD 适于分配给不同主机。当前 MLD 仅针对 Type 3 LD 进行架构定义。

MLD Port	An MLD Port is one that has linked up with an MLD Component. The port is natively bound to an FM- owned PPB inside the switch.	已与 MLD 组件建链的端口。该端口在交换机内部原生绑定到 FM 拥有的 PPB。
MMIO	Memory Mapped I/O	内存映射 I/O (Memory Mapped I/O)
MR	Multi-Root	多根 (Multi-Root)
MRL	Manually-operated Retention Latch as defined in PCIe Base Specification	手动操作保持门锁，如 PCIe 基础规范所定义
MS0	Meta0-State	Meta0-State
MSB	Most Significant Bit	最高有效位 (Most Significant Bit)
MSE	Memory Space Enable	内存空间使能 (Memory Space Enable)
MSI/MSI-X	Message Signaled Interrupt as defined in PCIe Base Specification	消息信号中断，如 PCIe 基础规范所定义
N/A	Not Applicable	不适用 (Not Applicable)
Native Width	This is the maximum possible expected negotiated link width.	可能协商得到的最大期望链路宽度。
NDR	No Data Response	无数据响应 (No Data Response)
NG	Number of Groups	组数 (Number of Groups)
NIB	Number of Bitmap Entries	位图条目数 (Number of Bitmap Entries)
NIW	Number of Interleave Ways	交织路数 (Number of Interleave Ways)
NOP	No Operation	空操作 (No Operation)
NT	Non-Temporal	非时间性 (Non-Temporal)
NVM	Non-volatile Memory	非易失性内存 (Non-volatile Memory)
NXM	Non-existent Memory	不存在的内存 (Non-existent Memory)

OBBF	Optimized Buffer Flush/Fill as defined in PCIe Base Specification	优化缓冲刷新/填充，如 PCIe 基础规范所定义
OHC	Orthogonal Header Content as defined in PCIe Base Specification	正交头内容，如 PCIe 基础规范所定义
OOB	Out of Band	带外（Out of Band）
_OSC	Operating System Capabilities as defined in ACPI Specification	操作系统能力，如 ACPI 规范所定义
OSCKID	CKID memory encryption key configured for use with non-TEE data.	配置用于非 TEE 数据的 CKID 内存加密密钥。
OSPM	Operating System-directed configuration and Power Management as defined in ACPI Specification	操作系统导向的配置与电源管理，如 ACPI 规范所定义
PA	Physical Address	物理地址（Physical Address）
P2P	Peer to peer	点对点 / 端到端（Peer to peer）
PBR	Port Based Routing	基于端口的路由（Port Based Routing）
PBR link	Link where all flits are in PBR format	所有 flit 均为 PBR 格式的链路
PBR switch	Switch that supports PBR and has PBR enabled	支持 PBR 且已启用 PBR 的交换机
P/C	Producer-Consumer	生产者-消费者（Producer-Consumer）
PCIe	PCI Express	PCI Express
PCRC	CRC-32 calculated on the flit plaintext content. Encrypted PCRC is used to provide robustness against hard and soft faults internal to the encryption and decryption engines.	对 flit 明文内容计算的 CRC-32。加密后的 PCRC 用于抵御加密与解密引擎内部的硬故障与软故障。
PDM	Private Device memory. Device-attached memory that is not mapped to system address space or directly accessible to Host as cacheable memory. Memory located on PCIe devices is of this type. Memory located on a CXL device can be mapped as either PDM or HDM.	私有设备内存（Private Device memory）。未映射到系统地址空间、也不可作为可缓存内存被主机直接访问的设备附属内存。PCIe 设备上的内存属于此类。CXL 设备上的内存可映射为 PDM 或 HDM。
Peer	Peer device in the context of peer-to-peer (P2P) accesses between devices.	在设备间点对点（P2P）访问上下文中的对端设备。
Persistent Memory Device	Device that retains content across power cycling. A CXL Memory device can advertise “Persistent Memory” capability as long as it supports the minimum set of requirements described in Section 8.2.10.9. The platform owns the final responsibility of determining whether a memory device can be used as Persistent Memory. This determination is beyond the scope of CXL specification.	断电后仍保留内容的设备。只要 CXL 内存设备支持第 8.2.10.9 节所述的最低要求集，即可通告“持久内存”能力。平台最终负责判定某内存设备是否可用作持久内存。该判定超出 CXL 规范范围。
PI	Programming Interface as defined in PCIe Base Specification	编程接口，如 PCIe 基础规范所定义
PID	PBR ID	PBR ID
PIF	PID Forward	PID 转发（PID Forward）
PLDM	Platform-Level Data Model	平台级数据模型（Platform-Level Data Model）
PM	Power Management	电源管理（Power Management）
PME	Power Management Event	电源管理事件（Power Management Event）
PM2IP	(Master) Power Manager to Independent Power Manager, PM messages from the host to the device.	（主）电源管理器到独立电源管理器，从主机到设备的 PM 消息。
PPB	PCI-to-PCI Bridge inside a CXL switch that is FM-owned. The port connected to a PPB can be disconnected, or connected to a PCIe component or connected to a CXL component.	CXL 交换机中由 FM 拥有的 PCI 到 PCI 桥。连接到 PPB 的端口可断开，或连接到 PCIe 组件，或连接到 CXL 组件。

PPR	Post Package Repair	封装后修复 (Post Package Repair)
PrimarySession	CMA/SPDM session established between the TSM or TSM RoT and the DSM. Utilized for configuring and locking the device and setting and clearing memory encryption keys using TSP.	在 TSM 或 TSM RoT 与 DSM 之间建立的 CMA/SPDM 会话。用于配置并锁定设备，以及使用 TSP 设置与清除内存加密密钥。
PSK	CMA/SPDM-defined pre-shared key.	CMA/SPDM 定义的预共享密钥。
PTH	PBR TLP Header	PBR TLP 头 (PBR TLP Header)
QoS	Quality of Service	服务质量 (Quality of Service)
QTG	QoS Throttling Group, the group of CXL.mem target resources that are throttled together in response to QoS telemetry (see Section 3.3.3). Each QTG is identified using a number that is known as QTG ID. QTG ID is a positive integer.	QoS 节流组 (QoS Throttling Group)，响应 QoS 遥测而一起被节流的一组 CXL.mem 目标资源 (见第 3.3.3 节)。每个 QTG 用称为 QTG ID 的正整数标识。
Rank	Set of memory devices on a channel that together execute a transaction.	通道上一组共同执行事务的内存设备。
RAS	Reliability Availability and Serviceability	可靠性、可用性与可服务性 (Reliability Availability and Serviceability)
RC	Root Complex	根复合体 (Root Complex)
RCD	Shorthand for a Device that is operating in RCD mode (formerly known as CXL 1.1 Device)	以 RCD 模式运行的设备的简称 (原称 CXL 1.1 设备)
RCD mode	Restricted CXL Device mode (formerly known as CXL 1.1 mode). The CXL operating mode with a number of restrictions. These restrictions include lack of Hot-Plug support and 68B Flit mode being the only supported flit mode. See Section 9.11.1 for the complete list of these restrictions.	受限 CXL 设备模式 (原称 CXL 1.1 模式)。带有若干限制的 CXL 运行模式。这些限制包括不支持热插拔，以及仅支持 68B Flit 模式。完整限制列表见第 9.11.1 节。
RCEC	Root Complex Event Collector, collects errors from PCIe RCiEPs, as defined in PCIe Base Specification.	根复合体事件收集器，收集来自 PCIe RCiEP 的错误，如 PCIe 基础规范所定义。
RCH	Restricted CXL Host. CXL Host that is operating in RCD mode (formerly known as CXL 1.1 Host).	受限 CXL 主机。以 RCD 模式运行的 CXL 主机 (原称 CXL 1.1 主机)。
RCiEP	Root Complex Integrated Endpoint	根复合体集成端点 (Root Complex Integrated Endpoint)
RCRB	Root Complex Register Block as defined in PCIe Base Specification	根复合体寄存器块，如 PCIe 基础规范所定义

RCS	Requestor Cache State – The cache coherency state tracked by the host or initiator	请求方缓存状态（Requestor Cache State）——由主机或发起方跟踪的缓存一致性状态
RDPAS	RCEC Downstream Port Association Structure	RCEC 下游端口关联结构（RCEC Downstream Port Association Structure）
Reserved	The contents, states, or information are not defined at this time. Reserved register fields must be read only and must return 0 (all 0s for multi-bit fields) when read. Reserved encodings for register and packet fields must not be used. Any implementation dependent on a Reserved field value or encoding will result in an implementation that is not CXL-spec compliant. The functionality of such an implementation cannot be guaranteed in this or any future revision of this specification. Flit, Slot, and message reserved bits should be cleared to 0 by the sender and the receiver should ignore them.	保留：内容、状态或信息此时尚未定义。保留寄存器字段必须为只读，读时必须返回 0（多位字段为全 0）。不得使用寄存器与包字段的保留编码。任何依赖保留字段值或编码的实现将导致不符合 CXL 规范；在本规范此版本或任何未来版本中均不能保证其功能。Flit、Slot 与消息的保留位应由发送方清零为 0，接收方应忽略它们。
RFM	Refresh Management	刷新管理（Refresh Management）
RGT	Routing Group Table	路由组表（Routing Group Table）
RP	Root Port	根端口（Root Port）
RPID	Requester PID. In the context of a GFD, the SPID from an incoming request.	请求方 PID。在 GFD 上下文中，指入站请求的 SPID。
RRS	Request Retry Status	请求重试状态（Request Retry Status）
RRSM	Remote Retry State Machine	远程重试状态机（Remote Retry State Machine）
RSDT	Root System Description Table as defined in ACPI Specification	根系统描述表，如 ACPI 规范所定义
RSVD or RV	Reserved	保留（Reserved）
RT	Route Table	路由表（Route Table）
RTT	Round-Trip Time	往返时延（Round-Trip Time）
RwD	Request with Data	带数据请求（Request with Data）
S2M	Subordinate to Master	从到主（Subordinate to Master）
SAT	SPID Access Table	SPID 访问表（SPID Access Table）
SBR	Secondary Bus Reset as defined in PCIe Base Specification	次级总线复位，如 PCIe 基础规范所定义
SCC	Sub-Class Code as defined in PCIe Base Specification	子类代码，如 PCIe 基础规范所定义
SDC	Silent Data Corruption	静默数据损坏（Silent Data Corruption）
SDS	Start of Data Stream	数据流开始（Start of Data Stream）
SecondarySession	One or more optional CMA/SPDM sessions established between a host entity and the DSM. Utilized for setting and clearing memory encryption keys using TSP.	在主机实体与 DSM 之间建立的一个或多个可选 CMA/SPDM 会话。用于使用 TSP 设置与清除内存加密密钥。
Selective Trust	Security model in which each TEE selects which devices it shall include in its TCB. The device trusted by one TEE may not be trusted by other TEEs within the system.	选择性信任：一种安全模型，每个 TEE 选择将哪些设备纳入其 TCB。某一 TEE 信任的设备可能不被系统中其他 TEE 信任。
SF	Snoop Filter	探听过滤器（Snoop Filter）
SFSC	Security Features for SCSI Commands	SCSI 命令安全特性（Security Features for SCSI Commands）
Sharer	Entity that is sharing data with another entity.	与另一实体共享数据的实体。
SHDA	Single Host, Direct Attached SLD EP	单主机、直连 SLD 端点
SH-MLD	Single headed MLD. CXL component that contains a single CXL port, presenting an MLD.	单头 MLD。包含单个 CXL 端口并呈现 MLD 的 CXL 组件。

SH-SLD	Single Headed Single Logical Device CXL component that contains a single CXL port, presenting an SLD.	单头单逻辑设备：包含单个 CXL 端口并呈现 SLD 的 CXL 组件。
SHSW	Single Host, Switch Attached SLD EP	单主机、交换机连接的 SLD 端点
SHSW-FM	Single Host, Fabric Managed, Switch Attached SLD EP	单主机、结构管理、交换机连接的 SLD 端点
Sideband	Signal used for device detection, configuration, and Hot-Plug in PCIe connectors, as defined in PCIe Base Specification	用于 PCIe 连接器中设备检测、配置与热插拔的信号，如 PCIe 基础规范所定义
SLD	Single Logical Device	单逻辑设备（Single Logical Device）
Smart I/O	Enhanced I/O with additional protocol support.	具有额外协议支持的增强 I/O。
SP	Security Protocol	安全协议（Security Protocol）
SPDM	Security Protocol and Data Model	安全协议与数据模型（Security Protocol and Data Model）
SPDM over DOE	Security Protocol and Data Model over Data Object Exchange as defined in PCIe Base Specification	基于数据对象交换的安全协议与数据模型，如 PCIe 基础规范所定义
SPID	Source PID	源 PID（Source PID）
sPPR	Soft Post Package Repair	软封装后修复（Soft Post Package Repair）
SRAT	System Resource Affinity Table as defined in ACPI Specification	系统资源亲和表，如 ACPI 规范所定义
SRIS	Separate Reference Clocks with Independent Spread Spectrum Clocking as defined in PCIe Base Specification	具有独立扩频时钟的分离参考时钟，如 PCIe 基础规范所定义

SV	Secret Value	秘密值 (Secret Value)
SVM	Shared Virtual Memory	共享虚拟内存 (Shared Virtual Memory)
SW	Software	软件 (Software)
Target	Defined by TSP as a memory expander device that receives TSP requests.	由 TSP 定义为接收 TSP 请求的内存扩展设备。
TC	Traffic Class	流量类别 (Traffic Class)
TCB	Trusted Computing Base - refers to the set of hardware, software and/or firmware entities that security assurances depend upon	可信计算基 (Trusted Computing Base) —— 安全保证所依赖的一组硬件、软件和/或固件实体
TDISP	PCI-SIG-defined TEE Device Interface Security Protocol	PCI-SIG 定义的 TEE 设备接口安全协议
TEE	Trusted Execution Environment	可信执行环境 (Trusted Execution Environment)
TEE Intent	TEE or non-TEE intent of a memory request from an initiator defined by the opcodes utilized in the transaction.	发起方内存请求的 TEE 或非 TEE 意图，由事务中使用的操作码定义。
TE State	TEE Exclusive State maintained by the device for each page or cacheline to enforce access control.	设备为每个页或缓存行维护的 TEE 独占状态，用于强制访问控制。
TLP	Transaction Layer Packet as defined in PCIe Base Specification	事务层包，如 PCIe 基础规范所定义
TMAC	Truncated Message Authentication Code	截断消息认证码 (Truncated Message Authentication Code)
TRP	Trailer Present	存在尾部 (Trailer Present)
TSM	TEE Security Manager. Logical entity within a host processor that is the TCB for a TVM and enforces security policies on the host.	TEE 安全管理器。主机处理器内作为 TVM 的 TCB 并在主机上强制执行安全策略的逻辑实体。
TSM RoT	TEE Security Manager Root of Trust. The entity on the host that establishes the CMA/SPDM PrimarySession.	TEE 安全管理器信任根。主机上建立 CMA/SPDM PrimarySession 的实体。
TSP	CXL-defined TEE Security Protocol. The collection of requirements and interfaces that allow memory devices to be utilized for confidential computing.	CXL 定义的 TEE 安全协议。允许内存设备用于机密计算的一组要求与接口。
TVM	TEE Virtual Machine. A TEE that has the property of a virtual machine. A TVM does not need to trust the hypervisor that hosts the TVM.	TEE 虚拟机。具有虚拟机属性的 TEE。TVM 无需信任托管该 TVM 的虚拟机监控器。
TVMCKID	CKID memory encryption key configured for use with TEE data.	配置用于 TEE 数据的 CKID 内存加密密钥。
UEFI	Unified Extensible Firmware Interface	统一可扩展固件接口 (Unified Extensible Firmware Interface)
UIE	Uncorrectable Internal Error	不可纠正内部错误 (Uncorrectable Internal Error)
UIG	Upstream Interleave Granularity	上游交织粒度 (Upstream Interleave Granularity)
UIO	Unordered Input/Output	无序输入/输出 (Unordered Input/Output)
UIW	Upstream Interleave Ways	上游交织路数 (Upstream Interleave Ways)
Upstream Port	Physical port that can be an upstream switch port, or an Endpoint port, or an RCD Upstream Port.	物理端口，可以是上游交换机端口、端点端口或 RCD 上游端口。
UQID	Unique Queue ID	唯一队列 ID (Unique Queue ID)
UR	Unsupported Request	不支持的请求 (Unsupported Request)
USAR	Upstream Acceptance Rules	上游接受规则 (Upstream Acceptance Rules)
USP	Upstream Switch Port	上游交换机端口 (Upstream Switch Port)
UTC	Coordinated Universal Time	协调世界时 (Coordinated Universal Time)

UUID	Universally Unique Identifier as defined in the IETF RFC 4122 Specification	通用唯一标识符，如 IETF RFC 4122 规范所定义
VA	Virtual Address	虚拟地址（Virtual Address）
VC	Virtual Channel	虚拟通道（Virtual Channel）
VCS	Virtual CXL Switch. Includes entities within the physical switch belonging to a single VH. It is identified using the VCS ID.	虚拟 CXL 交换机。包括属于单个 VH 的物理交换机内实体。使用 VCS ID 标识。
VDM	Vendor Defined Message	厂商定义消息（Vendor Defined Message）
vDSP	Downstream vPPB in a Host ES that is bound to one vUSP within a specific Downstream ES.	Host ES 中绑定到特定下游 ES 内某个 vUSP 的下游 vPPB。
VH	Virtual Hierarchy. Everything from the CXL RP down, including the CXL RP, CXL PPBs, and CXL Endpoints. Hierarchy ID means the same as PCIe.	虚拟层级。从 CXL RP 向下的一切，包括 CXL RP、CXL PPB 与 CXL 端点。Hierarchy ID 含义与 PCIe 相同。
VH Mode	A mode of operation where CXL RP is the root of the hierarchy	以 CXL RP 为层级根的运行模式
VID	Vendor ID	厂商 ID（Vendor ID）
vLSM	Virtual Link State Machine	虚拟链路状态机（Virtual Link State Machine）
VM	Virtual Machine	虚拟机（Virtual Machine）

VMM	Virtual Machine Manager	虚拟机监控器 (Virtual Machine Manager)
vPPB	Virtual PCI-to-PCI Bridge inside a CXL switch that is host-owned. Can be bound to a port that is either disconnected, connected to a PCIe component or connected to a CXL component.	CXL 交换机中由主机拥有的虚拟 PCI 到 PCI 桥。可绑定到断开、连接到 PCIe 组件或连接到 CXL 组件的端口。
VTV	VendPrefixL0 Target Vector	VendPrefixL0 目标向量 (VendPrefixL0 Target Vector)
vUSP	Upstream vPPB in a Downstream ES that is bound to one vDSP within a specific Host ES.	下游 ES 中绑定到特定 Host ES 内某个 vDSP 的上游 vPPB。
Warm Reset	As defined in PCIe Base Specification	热/温复位，如 PCIe 基础规范所定义
XSDT	Extended System Description Table as defined in ACPI Specification	扩展系统描述表，如 ACPI 规范所定义

1.3 Reference Documents / 参考文档

Table 1-2. Reference Documents / 表 1-2 参考文档

PCI Express Base Specification Revision 6.2 (abbreviated as PCIe Base Specification) PCI Express 基础规范修订版 6.2 (在 CXL 规范中简称为 PCIe 基础规范)	N/A	www.pcisig.com
PCI Firmware Specification (revision 3.3 or later) PCI 固件规范 (修订版 3.3 或更高)	Various / 多处	www.pcisig.com
Unordered IO (UIO) ECN to PCI Express Base Specification Revision 6.0 针对 PCI Express 基础规范修订版 6.0 的无序 I/O (UIO) ECN	N/A	members.pcisig.com/wg/PCI-SIG/document/19388
Management Message Passthrough via MMIO Mailbox (MMPT) ECN to PCI Express Specification Revision 6.1 针对 PCI Express 规范修订版 6.1 的经由 MMIO 邮箱的管理消息透传 (MMPT) ECN	N/A	members.pcisig.com/wg/PCI-SIG/document/20109
ACPI Specification (version 6.5 or later) ACPI 规范 (版本 6.5 或更高)	Various / 多处	www.uefi.org
Coherent Device Attribute Table (CDAT) Specification (version 1.04 or later) 一致性设备属性表 (CDAT) 规范 (版本 1.04 或更高)	Various / 多处	www.uefi.org/acpi
RFC 4122 A Universally Unique Identifier UUID URN Namespace RFC 4122 通用唯一标识符 UUID URN 命名空间	Various / 多处	www.ietf.org/rfc/rfc4122
UEFI Specification (version 2.10 or later) UEFI 规范 (版本 2.10 或更高)	Various / 多处	www.uefi.org
CXL Fabric Manager API over MCTP Binding Specification (DSP0234) (version 1.0.0 or later) 基于 MCTP 绑定的 CXL 结构管理器 API 规范 (DSP0234) (版本 1.0.0 或更高)	Chapter 7 / 第 7 章	www.dmtf.org/dsp/DSP0234
Management Component Transport Protocol (MCTP) Base Specification (DSP0236) (version 1.3.1 or later) 管理组件传输协议 (MCTP) 基础规范 (DSP0236) (版本 1.3.1 或更高)	Chapters 7, 8, 11 / 第 7、8、11 章	www.dmtf.org/dsp/DSP0236
MCTP SMBus/I2C Transport Binding Specification (DSP0237) (version 1.2.0 or later) MCTP SMBus/I2C 传输绑定规范 (DSP0237) (版本 1.2.0 或更高)	Chapter 11 / 第 11 章	www.dmtf.org/dsp/DSP0237
MCTP PCIe VDM Transport Binding Specification (DSP0238) (version 1.2.0 or later) MCTP PCIe VDM 传输绑定规范 (DSP0238) (版本 1.2.0 或更高)	Chapters 7, 11 / 第 7、11 章	www.dmtf.org/dsp/DSP0238
Security Protocol and Data Model (SPDM) Specification (DSP0274) (version 1.2.0 or later) 安全协议与数据模型 (SPDM) 规范 (DSP0274) (版本 1.2.0 或更高)	Chapters 11, 14 / 第 11、14 章	www.dmtf.org/dsp/DSP0274
SPDM over MCTP Binding Specification (DSP0275) (version 1.0.0 or later) 基于 MCTP 绑定的 SPDM 规范 (DSP0275) (版本 1.0.0 或更高)	Chapter 11 / 第 11 章	www.dmtf.org/dsp/DSP0275
Secured Messages using SPDM over MCTP Binding Specification (DSP0276) (version 1.0.0 or later) 基于 MCTP 绑定使用 SPDM 的安全消息规范 (DSP0276) (版本 1.0.0 或更高)	Chapter 11 / 第 11 章	www.dmtf.org/dsp/DSP0276
Secured Messages using SPDM Specification (DSP0277) (version 1.0.0 or later) 使用 SPDM 的安全消息规范 (DSP0277) (版本 1.0.0 或更高)	Chapter 11 / 第 11 章	www.dmtf.org/dsp/DSP0277
CXL Type 3 Component Command Interface over MCTP Binding Specification (DSP0281) (version 1.0.0 or later) 基于 MCTP 绑定的 CXL Type 3 组件命令接口规范 (DSP0281) (版本 1.0.0 或更高)	Chapters 7, 9 / 第 7、9 章	www.dmtf.org/dsp/DSP0281
NIST SP 800-38D Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC NIST SP 800-38D 分组密码工作模式建议: 伽罗瓦/计数器模式 (GCM) 与 GMAC	Chapters 8, 11 / 第 8、11 章	nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38d.pdf
JEDEC DDR5 Specification, JESD79-5 (5B, version 1.2 or later) JEDEC DDR5 规范, JESD79-5 (5B, 版本 1.2 或更高)	Chapters 8, 13 / 第 8、13 章	www.jedec.org
Security Features for SCSI Commands (SFSC) SCSI 命令安全特性 (SFSC)	Chapter 8 / 第 8 章	webstore.ansi.org

Note 1: Peer-to-peer flows to CXL.mem regions are supported using Unordered I/O (UIO) on the CXL.io protocol or with Direct CXL.mem access as described in this specification.

中文

注 1：到 CXL.mem 区域的点对点流可通过 CXL.io 协议上的无序 I/O (UIO)，或通过本规范所述的直接 CXL.mem 访问来支持。

1.4 Motivation and Overview / 动机与概述

1.4.1 CXL

EN
CXL is a dynamic multi-protocol technology designed to support accelerators and memory devices. CXL provides a rich set of protocols that include I/O semantics similar to PCIe (i.e., CXL.io), caching protocol semantics (i.e., CXL.cache), and memory access semantics (i.e., CXL.mem) over a discrete or on-package link. CXL.io is required for discovery and enumeration, error reporting, P2P accesses to CXL memory and host physical address (HPA) lookup. CXL.cache and CXL.mem protocols may be optionally implemented by the particular accelerator or memory device usage model. An important benefit of CXL is that it provides a low-latency, high-bandwidth path for an accelerator to access the system and for the system to access the memory attached to the CXL device. Figure 1-1 is a conceptual diagram that shows a device attached to a Host processor via CXL.

中文
CXL 是一种面向加速器与内存设备的动态多协议技术。CXL 提供丰富的协议集合，包括与 PCIe 类似的 I/O 语义（即 CXL.io）、缓存协议语义（即 CXL.cache）以及内存访问语义（即 CXL.mem），可运行于分立链路或封装内链路上。CXL.io 是发现与枚举、错误报告、对 CXL 内存的 P2P 访问以及主机物理地址（HPA）查找所必需的。CXL.cache 与 CXL.mem 协议可由特定加速器或内存设备使用模型选择实现。CXL 的一项重要收益是：它为加速器访问系统、以及系统访问连接到 CXL 设备的内存，提供了低延迟、高带宽路径。图 1-1 是经由 CXL 将设备连接到主机处理器的概念图。

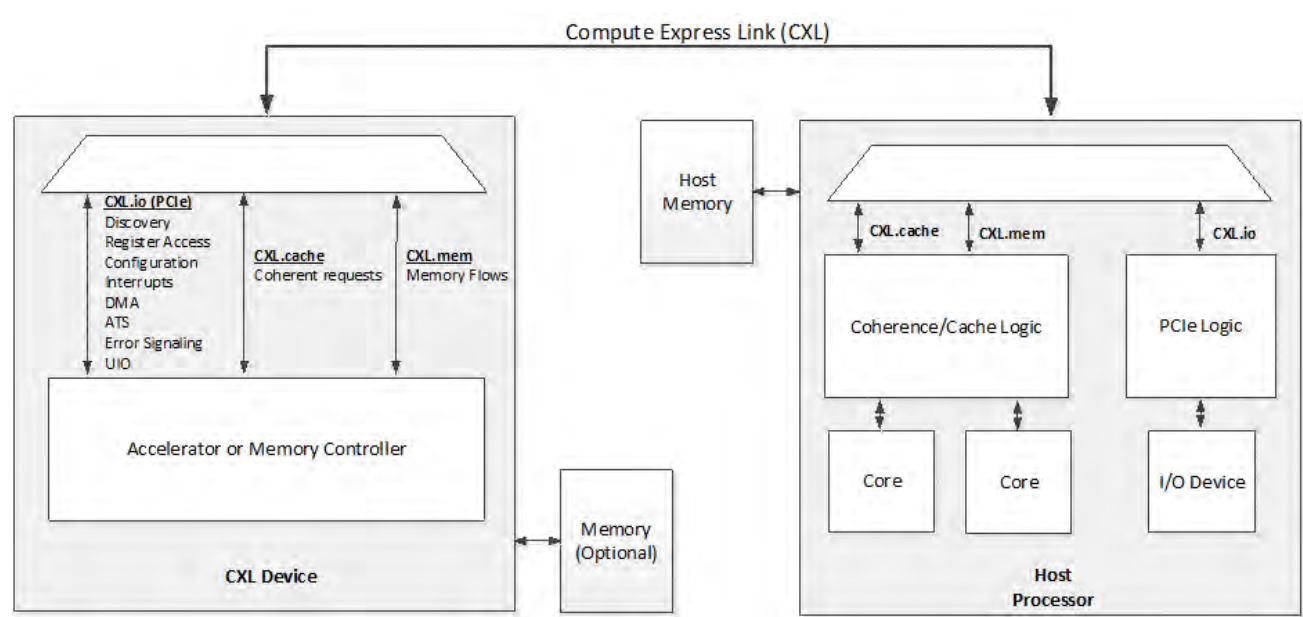


Figure 1-1. Conceptual Diagram of Device Attached to Processor via CXL / 图 1-1. 设备经 CXL 连接到处理器的概念图

EN
The CXL 2.0 specification enables additional usage models beyond the CXL 1.1 specification, while being fully backward compatible with the CXL 1.1 (and CXL 1.0) specification. It enables managed Hot-Plug, security enhancements, persistent memory support, memory error reporting, and telemetry. The CXL 2.0 specification also enables single-level switching support for fan-out as well as the ability to pool devices across multiple virtual hierarchies, including multi-domain support of memory devices. Figure 1-2 demonstrates memory and accelerator disaggregation through single level switching, in addition to fan-out, across multiple virtual hierarchies, each represented by a unique color. The CXL 2.0 specification also enables these resources (memory or accelerators) to be off-lined from one domain and on-lined into another domain, thereby allowing the resources to be time-multiplexed across different virtual hierarchies, depending on their resource demand.

中文
CXL 2.0 规范在完全向后兼容 CXL 1.1（以及 CXL 1.0）规范的同时，支持超出 CXL 1.1 的更多使用模型。它支持受管热插拔、安全增强、持久内存支持、内存错误报告与遥测。CXL 2.0

还支持用于扇出的单级交换，以及跨多个虚拟层级池化设备的能力，包括内存设备的多域支持。图 1-2 展示了通过单级交换实现内存与加速器解聚，以及跨多个虚拟层级（各用不同颜色表示）的扇出。CXL 2.0 还允许将这些资源（内存或加速器）从一个域下线并上线到另一域，从而可按资源需求在不同虚拟层级间进行时分复用。

EN

The CXL 3.0 specification doubles the bandwidth while enabling additional usage models beyond the CXL 2.0 specification. The CXL 3.0 specification is fully backward compatible with the CXL 2.0 specification (and hence with the CXL 1.1 and CXL 1.0 specifications). The maximum Data Rate doubles to 64.0 GT/s with PAM-4 signaling, leveraging the PCIe Base Specification PHY along with its CRC and FEC, to double the bandwidth, with provision for an optional Flit arrangement for low latency. Multi-level switching is enabled with the CXL 3.0 specification, supporting up to 4K Ports, to enable CXL to evolve as a fabric extending, including non-tree topologies, to the Rack and Pod level. The CXL 3.0 specification enables devices to perform direct peer-to-peer accesses to HDM memory using UIO (in addition to MMIO memory that existed before) to deliver performance at scale, as shown in Figure 1-3. Snoop Filter support can be implemented in Type 2 and Type 3 devices to enable direct peer-to-peer accesses using the back-invalidate channels introduced in CXL.mem. Shared memory support across multiple virtual hierarchies is provided for collaborative processing across multiple virtual hierarchies, as shown in Figure 1-4.

中文

CXL 3.0 规范在带宽翻倍的同时，支持超出 CXL 2.0 的更多使用模型，并与 CXL 2.0（因此也与 CXL 1.1、CXL 1.0）完全向后兼容。最大数据速率在 PAM-4 信令下提升至 64.0 GT/s，利用 PCIe 基础规范 PHY 及其 CRC 与 FEC 实现带宽翻倍，并可选采用面向低延迟的 Flit 编排。CXL 3.0 启用多级交换，最多支持 4K 端口，使 CXL 可演进为扩展到机架与 Pod 级别的结构（含非树形拓扑）。CXL 3.0 允许设备使用 UIO 对 HDM 内存进行直接点对点访问（此外仍支持此前已有的 MMIO 内存），以实现规模化性能，如图 1-3 所示。Type 2 与 Type 3 设备可实现探测过滤器支持，以便利用 CXL.mem 引入的反向使无效通道进行直接点对点访问。还提供跨多个虚拟层级的共享内存支持，用于跨虚拟层级的协同处理，如图 1-4 所示。

EN

CXL protocol is compatible with PCIe CEM Form Factor (4.0 and later), all form factors relating to EDSFF SSF-TA-1009 (revision 2.0 and later) and other form factors that support PCIe.

中文

CXL 协议与 PCIe CEM 形态规格（4.0 及更高）、与 EDSFF SSF-TA-1009（修订版 2.0 及更高）相关的所有形态规格，以及其他支持 PCIe 的形态规格兼容。

Figure 1-2. Fan-out and Pooling Enabled by Switches / 图 1-2. 交换机实现的扇出与池化（原图为矢量示意，见源 PDF 第 63 页）

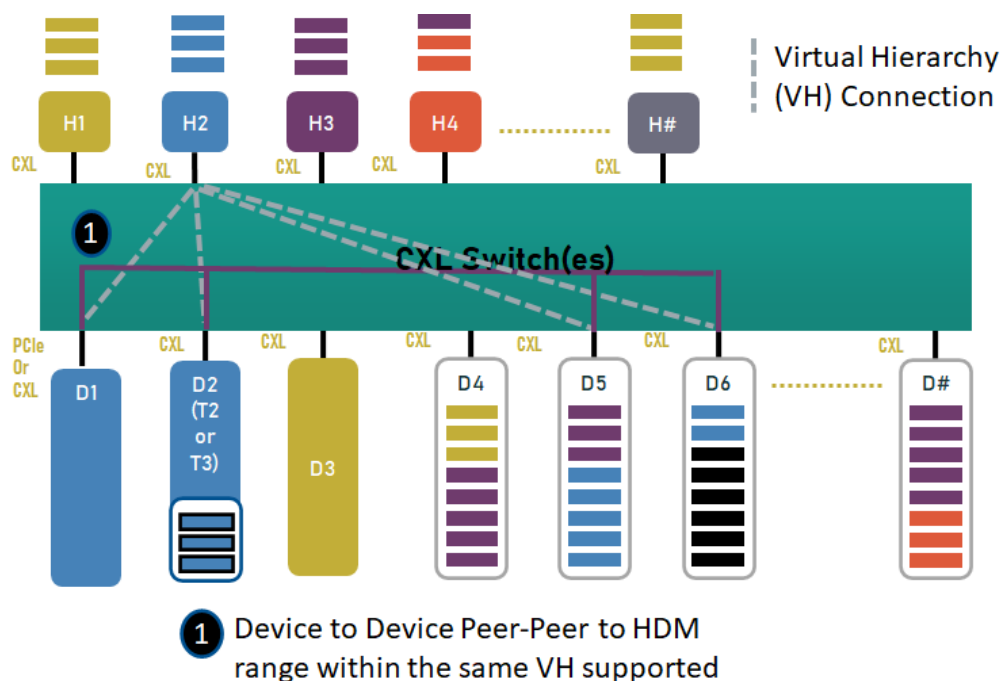


Figure 1-3. Direct Peer-to-Peer Access to an HDM Memory by PCIe/CXL Devices without Going through the Host / 图 1-3. PCIe/CXL 设备不经主机直接点对点访问 HDM 内存

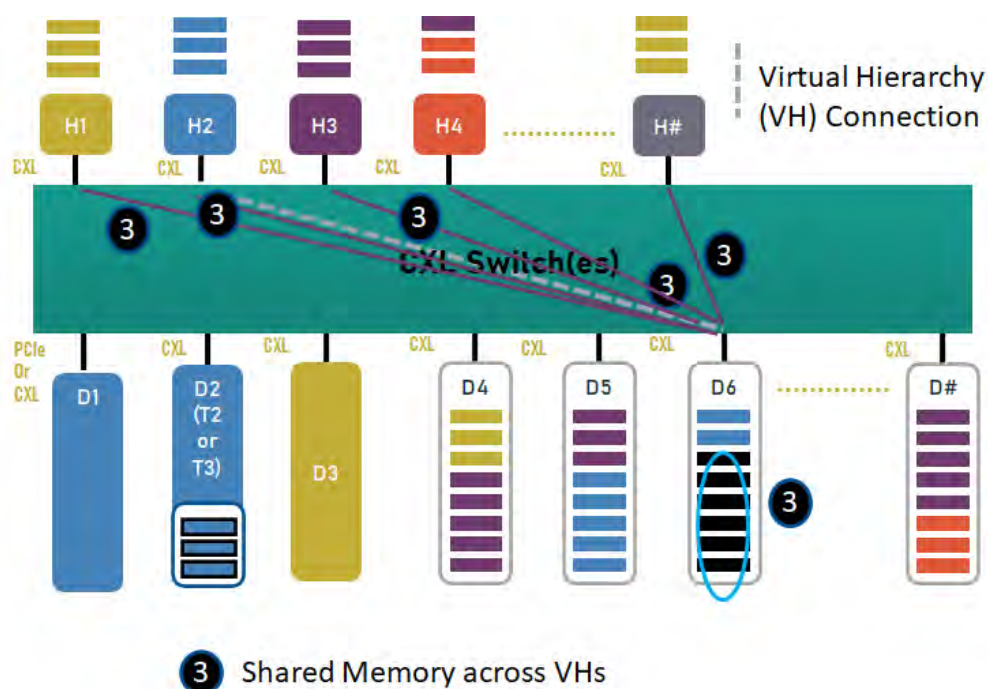


Figure 1-4. Shared Memory across Multiple Virtual Hierarchies / 图 1-4. 跨多个虚拟层级的共享内存

1.4.2 Flex Bus

EN

A Flex Bus port allows designs to choose between providing native PCIe protocol or CXL over a high-bandwidth, off-package link; the selection happens during link training via alternate protocol negotiation and depends on the device that is plugged into the slot. Flex Bus uses PCIe electricals, making it compatible with PCIe retimers and with form factors that support PCIe.

中文

Flex Bus 端口允许设计在高带宽、封装外链路上选择提供原生 PCIe 协议或 CXL；选择在链路训练期间通过备用协议协商完成，并取决于插入插槽的设备。Flex Bus 使用 PCIe 电气特性，因此与 PCIe 重定时器以及支持 PCIe 的形态规格兼容。

EN

Figure 1-5 provides a high-level diagram of a Flex Bus port implementation, illustrating both a slot implementation and a custom implementation where the device is soldered down on the motherboard. The slot implementation can accommodate either a Flex Bus.CXL card or a PCIe card. One or two optional retimers can be inserted between the CPU and the device to extend the channel length. As illustrated in Figure 1-6, this flexible port can be used to attach coherent accelerators or smart I/O to a Host processor. Figure 1-7 illustrates how a Flex Bus.CXL port can be used as a memory expansion port.

中文

图 1-5 给出 Flex Bus 端口实现的高层示意，同时展示插槽实现以及设备焊接在主板上的定制实现。插槽实现可容纳 Flex Bus.CXL 卡或 PCIe 卡。可在 CPU 与设备之间插入一或两个可选重定时器以延长通道。如图 1-6 所示，该灵活端口可用于将一致性加速器或智能 I/O 连接到主机处理器。图 1-7 说明如何将 Flex Bus.CXL 端口用作内存扩展端口。

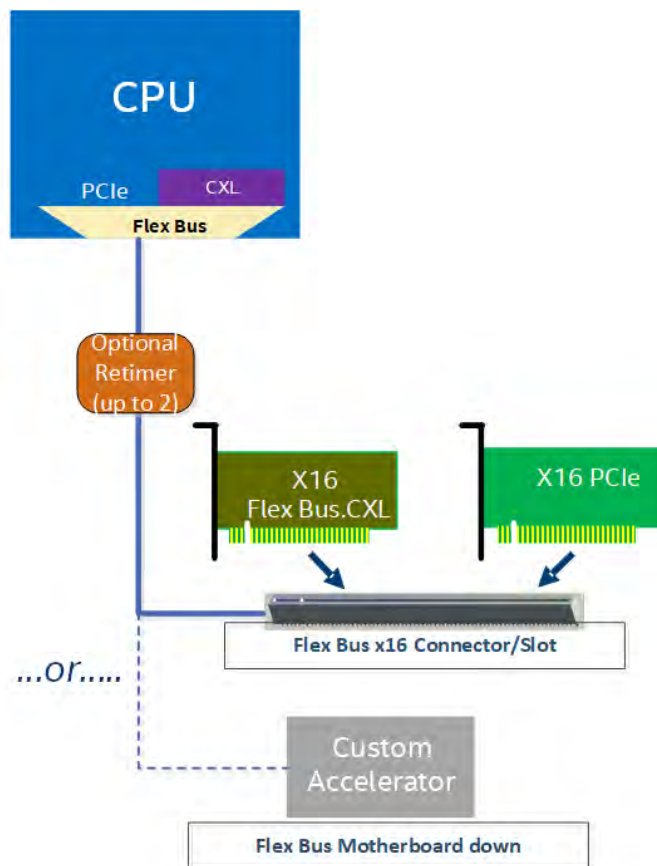


Figure 1-5. CPU Flex Bus Port Example / 图 1-5. CPU Flex Bus 端口示例

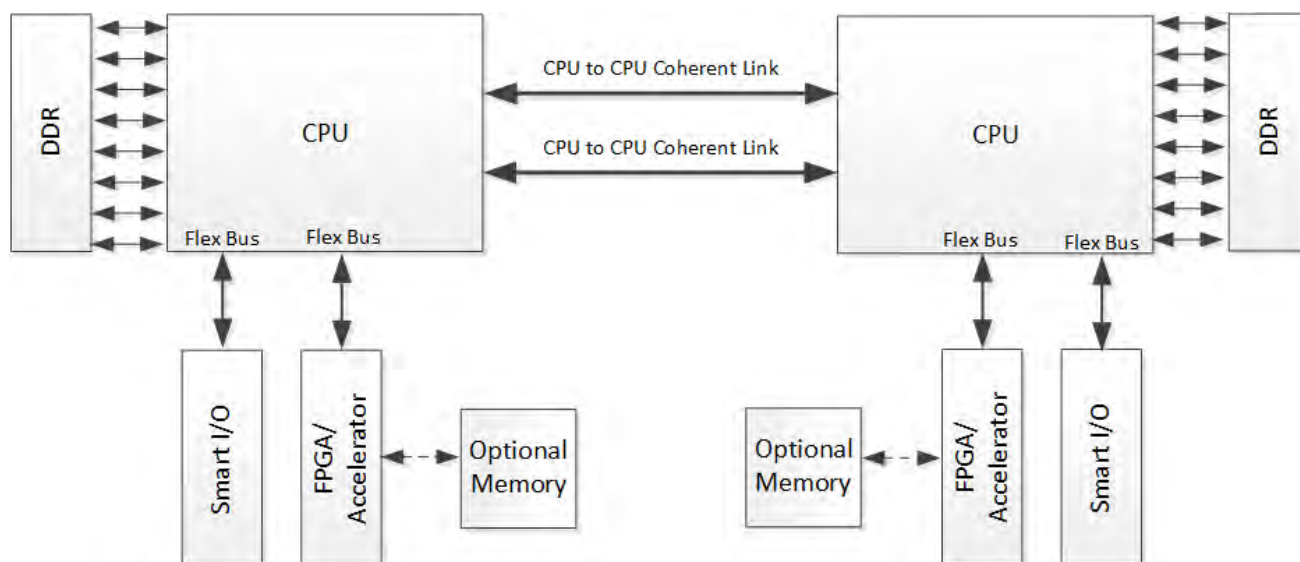


Figure 1-6. Flex Bus Usage Model Examples / 图 1-6. Flex Bus 使用模型示例

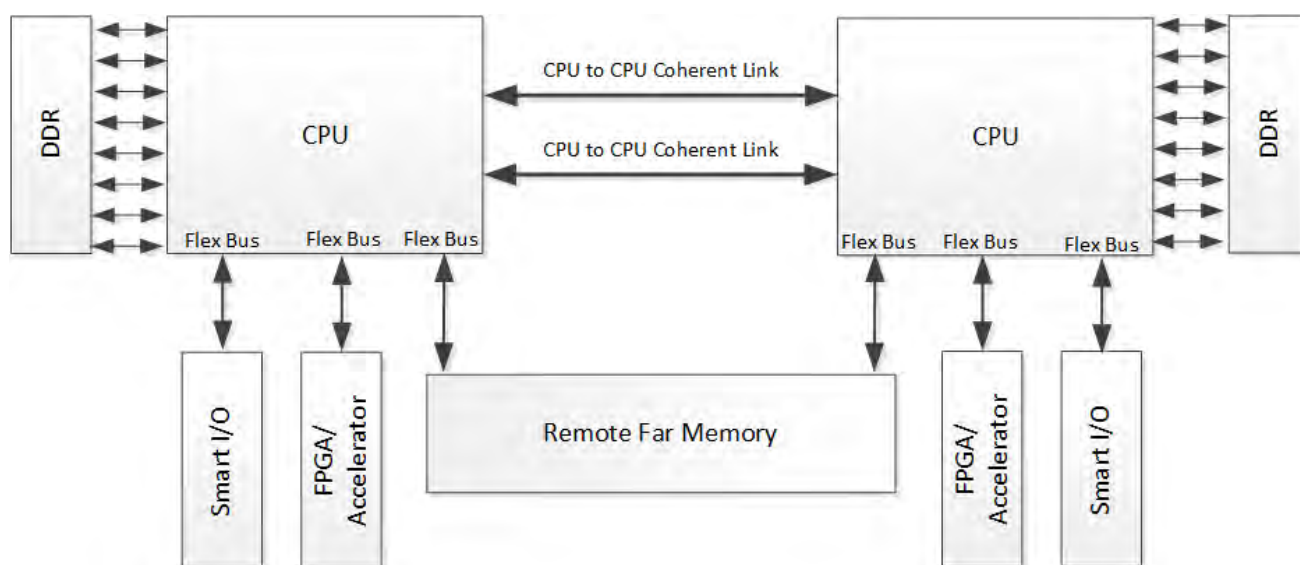


Figure 1-7. Remote Far Memory Usage Model Example / 图 1-7. 远程远端内存使用模型示例

EN

Figure 1-8 illustrates the connections that are supported below a CXL Downstream Port.

中文

图 1-8 说明 CXL 下游端口下方所支持的连接。

Figure 1-8. CXL Downstream Port Connections / 图 1-8. CXL 下游端口连接（示意见源 PDF 第 67 页）

1.5 Flex Bus Link Features / Flex Bus 链路特性

EN

Flex Bus provides a point-to-point interconnect that can transmit native PCIe protocol or dynamic multi-protocol CXL to provide I/O, caching, and memory protocols over PCIe electricals. The primary link attributes include support of the following features:

中文

Flex Bus 提供点对点互连，可传输原生 PCIe 协议或动态多协议 CXL，从而在 PCIe 电气上提供 I/O、缓存与内存协议。主要链路属性包括以下特性支持：

- Native PCIe mode, full feature support as defined in PCIe Base Specification
原生 PCIe 模式，完整特性支持如 PCIe 基础规范所定义
- CXL mode, as defined in this specification
CXL 模式，如本规范所定义
- Configuration of PCIe vs. CXL protocol mode
PCIe 与 CXL 协议模式的配置
- With PAM4, signaling rate of 64 GT/s, and degraded rates of 32 GT/s, 16 GT/s, or 8 GT/s in CXL mode. Otherwise, signaling rate of 32 GT/s, degraded rate of 16 GT/s or 8 GT/s in CXL mode
在 PAM4 下，CXL 模式信令速率为 64 GT/s，降级速率为 32/16/8 GT/s；否则 CXL 模式信令速率为 32 GT/s，降级速率为 16 或 8 GT/s
- Link width support for x16, x8, x4, x2 (degraded mode), and x1 (degraded mode) in CXL mode
CXL 模式下链路宽度支持 x16、x8、x4、x2（降级）与 x1（降级）
- Bifurcation (aka Link Subdivision) support to x4 in CXL mode
CXL 模式下支持分叉（即链路细分）至 x4

1.6 Flex Bus Layering Overview / Flex Bus 分层概述

EN

Flex Bus architecture is organized as multiple layers, as illustrated in Figure 1-9. The CXL transaction (protocol) layer is subdivided into logic that handles CXL.io and logic that handles CXL.cache and CXL.mem; the CXL link layer is subdivided in the same manner. Note that the CXL.cache and CXL.mem logic are combined within the transaction layer and within the link layer. The CXL link layer interfaces with the CXL ARB/MUX, which interleaves the traffic from the two logic streams. Additionally, the PCIe transaction and data link layers are optionally implemented and, if implemented, are permitted to be converged with the CXL.io transaction and link layers, respectively. As a result of the link training process, the transaction and link layers are configured to operate in either PCIe mode or CXL mode. While a host CPU would most likely implement both modes, an accelerator AIC is permitted to implement only the CXL mode. The logical sub-block of the Flex Bus physical layer is a converged logical physical layer that can operate in either PCIe mode or CXL mode, depending on the results of alternate mode negotiation during link training.

中文

Flex Bus 架构组织为多层，如图 1-9 所示。CXL 事务（协议）层细分为处理 CXL.io 的逻辑，以及处理 CXL.cache 与 CXL.mem 的逻辑；CXL 链路层亦按同样方式细分。注意：CXL.cache 与 CXL.mem 逻辑在事务层与链路层内部是合并的。CXL 链路层与 CXL ARB/MUX 接口，后者对两路逻辑流的流量进行交织。此外，PCIe 事务层与数据链路层为可选实现；若实现，则允许分别与 CXL.io 事务层和链路层融合。链路训练结束后，事务层与链路层被配置为以 PCIe 模式或 CXL 模式运行。主机 CPU 很可能实现两种模式，而加速器 AIC 允许仅实现 CXL 模式。Flex Bus 物理层的逻辑子块是融合的逻辑物理层，可根据链路训练期间备用模式协商结果，以 PCIe 模式或 CXL 模式运行。

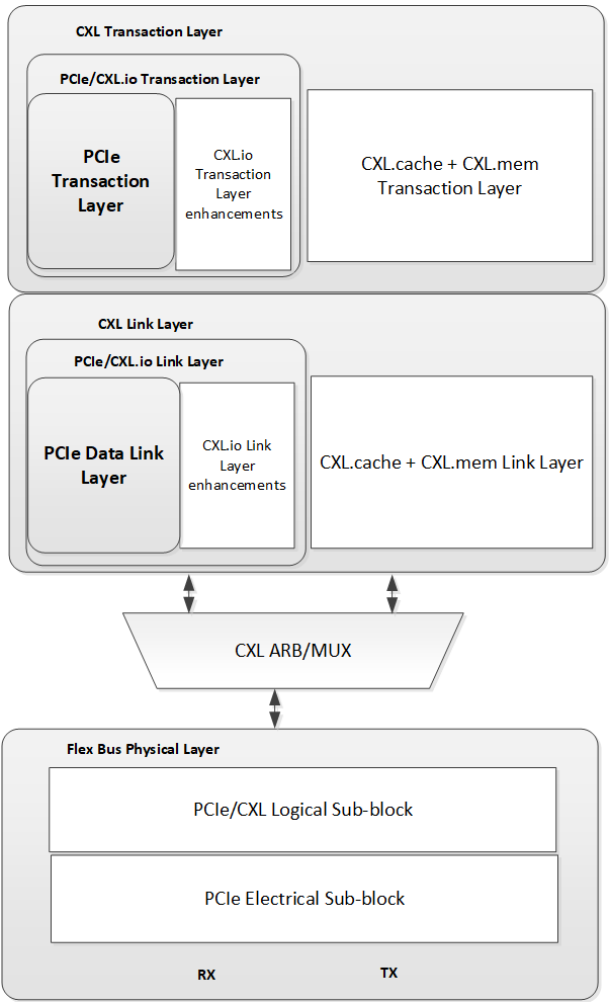


Figure 1-9. Conceptual Diagram of Flex Bus Layering / 图 1-9. Flex Bus 分层概念图

1.7 Document Scope / 文档范围

EN
This document specifies the functional and operational details of the Flex Bus interconnect and the CXL protocol. It describes the CXL usage model and defines how the transaction, link, and physical layers operate. Reset, power management, and initialization/configuration flows are described. Additionally, RAS behavior is described. Refer to PCIe Base Specification for PCIe protocol details.

中文
本文档规定 Flex Bus 互连与 CXL 协议的功能与运行细节。它描述 CXL 使用模型，并定义事务层、链路层与物理层如何工作。同时描述复位、电源管理以及初始化/配置流程，并描述 RAS 行为。PCIe 协议细节请参阅 PCIe 基础规范。

EN
The contents of this document are summarized in the following chapter highlights:

中文
本文档内容按以下章节要点概述：

- Chapter 2.0, “CXL System Architecture” – This chapter describes Type 1, Type 2, and Type 3 devices that might attach to a CPU Root Complex or a CXL switch over a CXL-capable link. For each device profile, a description of the typical workload and system resource usage is provided along with an explanation of which CXL capabilities are relevant for that workload. Bias-based and Back-Invalidate-based coherency models are introduced. This chapter also covers multi-headed devices and G-FAM devices and how they enable memory pooling and memory sharing usages. It also provides a summary of the CXL fabric extensions and its scalability.

第 2.0 章 “CXL 系统架构”——描述可能通过支持 CXL 的链路连接到 CPU 根复合体或 CXL 交换机的 Type 1、Type 2 与 Type 3 设备。对每种设备配置给出典型工作负载与系统资源使用说明，并解释哪些 CXL 能力与该工作负载相关。介绍基于 Bias 与基于 Back-Invalidate 的一致性模型。本章还涵盖多头设备与 G-FAM 设备，及其如何支持内存池化与内存共享，并概述 CXL 结构扩展及其可扩展性。

• Chapter 3.0, “CXL Transaction Layer” – This chapter is divided into subsections that describe details for CXL.io, CXL.cache, and CXL.mem. The CXL.io protocol is required for all implementations, while the other two protocols are optional depending on expected device usage and workload. The transaction layer specifies the transaction types, transaction layer packet formatting, transaction ordering rules, and crediting. The CXL.io protocol is based on the “Transaction Layer Specification” chapter of PCIe Base Specification; any deltas from PCIe Base Specification are described in this chapter. These deltas include PCIe Vendor Defined Messages for reset and power management, modifications to the PCIe ATS request and completion formats to support accelerators. For CXL.cache, this chapter describes the channels in each direction (i.e., request, response, and data), the transaction opcodes that flow through each channel, and the channel crediting and ordering rules. The transaction fields associated with each channel are also described. For CXL.mem, this chapter defines the message classes in each direction, the fields associated with each message class, and the message class ordering rules. Finally, this chapter provides flow diagrams that illustrate the sequence of transactions involved in completing host-initiated and device-initiated accesses to device-attached memory.

第 3.0 章 “CXL 事务层”——分节描述 CXL.io、CXL.cache 与 CXL.mem。所有实现均需 CXL.io；另两种协议可视预期设备用途与工作负载选择实现。事务层规定事务类型、事务层包格式、事务排序规则与信用机制。CXL.io 基于 PCIe 基础规范 “事务层规范” 章节；与 PCIe 的差异在本章说明，包括用于复位与电源管理的 PCIe 厂商定义消息，以及对 PCIe ATS 请求与完成格式的修改以支持加速器。对 CXL.cache，本章描述各方向通道（请求、响应与数据）、流经各通道的事务操作码，以及通道信用与排序规则，并描述各通道相关事务字段。对 CXL.mem，本章定义各方向消息类、各类相关字段以及消息类排序规则。最后给出流程图，说明完成主机发起与设备发起的对设备附属内存访问所涉及的事务序列。

• Chapter 4.0, “CXL Link Layers” – The link layer is responsible for reliable transmission of the transaction layer packets across the Flex Bus link. This chapter is divided into subsections that describe details for CXL.io and for CXL.cache and CXL.mem. The CXL.io protocol is based on the “Data Link Layer Specification” chapter of PCIe Base Specification; any deltas from PCIe Base Specification are described in this chapter. For CXL.cache and CXL.mem, the 68B flit format and 256B flit format are specified. The flit packing rules for selecting transactions from internal queues to fill the slots in the flit are described. Other features described for 68B flit mode include the retry mechanism, link layer control flits, CRC calculation, and viral and poison.

第 4.0 章 “CXL 链路层”——链路层负责在 Flex Bus 链路上可靠传输事务层包。本章分节描述 CXL.io 以及 CXL.cache 与 CXL.mem。CXL.io 基于 PCIe 基础规范 “数据链路层规范” 章节；差异在本章说明。对 CXL.cache 与 CXL.mem，规定 68B 与 256B flit 格式，并描述从内部队列选择事务填充 flit 槽位的打包规则。对 68B flit 模式还描述重试机制、链路层控制 flit、CRC 计算，以及 viral 与 poison。

• Chapter 5.0, “CXL ARB/MUX” – The ARB/MUX arbitrates between requests from the CXL link layers and multiplexes the data to forward to the physical layer. On the receiver side, the ARB/MUX decodes the flit to determine the target to forward transactions to the appropriate CXL link layer. Additionally, the ARB/MUX maintains virtual link state machines for every link layer it interfaces with, processing power state transition requests from the local link layers and generating ARB/MUX link management packets to communicate with the remote ARB/MUX.

第 5.0 章 “CXL ARB/MUX”——ARB/MUX 对来自 CXL

链路层的请求进行仲裁，并将数据复用后转发给物理层。在接收侧，ARB/MUX 解码 flit 以确定目标，并将事务转发到相应 CXL 链路层。此外，ARB/MUX 为其对接的每个链路层维护虚拟链路状态机，处理本地链路层的电源状态转换请求，并生成 ARB/MUX 链路管理包与远端 ARB/MUX 通信。

• Chapter 6.0, “Flex Bus Physical Layer” – The Flex Bus physical layer is responsible for training the link to bring it to operational state for transmission of PCIe packets or CXL flits. During operational state, it prepares the data from the CXL link layers or the PCIe link layer for transmission across the Flex Bus link; likewise, it converts data received from the link to the appropriate format to pass on to the appropriate link layer. This chapter describes the deltas from PCIe Base Specification to support the CXL mode of operation. The framing of the CXL flits and the physical layer packet layout for 68B Flit mode as well as 256B Flit mode are described. The mode selection process to decide between CXL mode or PCIe mode, including hardware autonomous negotiation and software-controlled selection is also described. Finally, CXL low-latency modes are described.

第 6.0 章 “Flex Bus 物理层”——负责训练链路使其进入可传输 PCIe 包或 CXL flit 的运行状态。在运行状态中，它准备来自 CXL 链路层或 PCIe 链路层的数据以便在 Flex Bus

链路上传输；同样，它将从链路接收的数据转换为适当格式并交给相应链路层。本章描述相对 PCIe 基础规范为支持 CXL 模式所需的差异，说明 68B 与 256B Flit 模式下 CXL flit 成帧与物理层包布局，并描述在 CXL 与 PCIe

模式间选择的过程（含硬件自主协商与软件控制选择），最后描述 CXL 低延迟模式。

- Chapter 7.0, “Switching” – This chapter provides an overview of different CXL switching configurations and describes rules for how to configure switches. Additionally, the Fabric Manager application interface is specified and CXL Fabric is introduced.

第 7.0 章“交换”——概述不同 CXL 交换配置并描述交换机配置规则。此外规定结构管理器应用接口，并引入 CXL Fabric。

- Chapter 8.0, “Control and Status Registers” – This chapter provides details of the Flex Bus and CXL control and status registers. It describes the configuration space and memory mapped registers that are located in various CXL components. It also describes the CXL Component Command Interface.

第 8.0 章“控制与状态寄存器”——详述 Flex Bus 与 CXL 控制与状态寄存器，描述位于各 CXL 组件中的配置空间与内存映射寄存器，并描述 CXL 组件命令接口。

- Chapter 9.0, “Reset, Initialization, Configuration, and Manageability” – This chapter describes the flows for boot, reset entry, and sleep-state entry; this includes the transactions sent across the link to initiate and acknowledge entry as well as steps taken by a CXL device to prepare for entry into each of these states. Additionally, this chapter describes the software enumeration model of both RCD and CXL virtual hierarchy and how the System Firmware view of the hierarchy may differ from the OS view. This chapter discusses the software view of CXL.mem, CXL extensions to Firmware-OS interfaces, and the CXL device manageability model.

第 9.0 章“复位、初始化、配置与可管理性”——描述启动、进入复位与进入睡眠状态的流程，包括链路上用于发起并确认进入的事务，以及 CXL 设备为进入各状态所做准备。此外描述 RCD 与 CXL

虚拟层级的软件枚举模型，以及系统固件与操作系统对层级视图可能存在的差异；并讨论 CXL.mem 的软件视图、对固件-OS 接口的 CXL 扩展，以及 CXL 设备可管理性模型。

- Chapter 10.0, “Power Management” – This chapter provides details on protocol specific link power management and physical layer power management. It describes the overall power management flow in three phases: protocol specific PM entry negotiation, PM entry negotiation for ARB/MUX interfaces (managed independently per protocol), and PM entry process for the physical layer. The PM entry process for CXL.cache and CXL.mem is slightly different than the process for CXL.io; these processes are described in separate subsections in this chapter.

第 10.0 章“电源管理”——详述协议特定的链路电源管理与物理层电源管理。整体电源管理流程分三阶段：协议特定的 PM 进入协商、ARB/MUX 接口的 PM 进入协商（按协议独立管理），以及物理层 PM 进入过程。CXL.cache 与 CXL.mem 的 PM 进入过程与 CXL.io 略有不同，分别在本章各小节描述。

- Chapter 11.0, “CXL Security” – This chapter provides details on the CXL Integrity and Data Encryption (CXL IDE) scheme that is used for securing CXL protocol flits that are transmitted across the link, IDE modes, and configuration flows.

第 11.0 章“CXL 安全”——详述用于保护链路上传输的 CXL 协议 flit 的 CXL 完整性与数据加密（CXL IDE）方案、IDE 模式与配置流程。

- Chapter 12.0, “Reliability, Availability, and Serviceability” – This chapter describes the RAS capabilities supported by a CXL host, a CXL switch, and a CXL device. It describes how various types of errors are logged and signaled to the appropriate hardware or software error handling agent. It describes the Link Down flow and the viral handling expectation. Finally, it describes the error injection requirements.

第 12.0 章“可靠性、可用性与可服务性”——描述 CXL 主机、交换机与设备支持的 RAS 能力；说明各类错误如何被记录并通告给相应硬件或软件错误处理代理；描述链路断开流程与 viral 处理期望；最后描述错误注入要求。

- Chapter 13.0, “Performance Considerations” – This chapter describes hardware and software considerations for optimizing performance across the Flex Bus link in CXL mode. It also describes the performance monitoring infrastructure for CXL components.

第 13.0 章“性能考虑”——描述在 CXL 模式下优化 Flex Bus 链路性能的硬件与软件考虑，并描述 CXL 组件的性能监控基础设施。

- Chapter 14.0, “CXL Compliance Testing” – This chapter describes methodologies for ensuring that a device is compliant with the CXL specification.

第 14.0 章“CXL 合规测试”——描述确保设备符合 CXL 规范的方法。

— 第 1 章结束 / End of Chapter 1 —

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